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Roll No

EE-8003 (3) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Power Quality
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Discuss the major reasons for the growing concern about the quality of Electric power by both electrical utilities and end users.
b) How Electrical transients are generated?
2. a) What is the need for protection against over voltages?
b) Explain the different methods of estimating sag performance.
3. a) What are the different voltage sag mitigation techniques? Explain in details.
b) Define the term voltage sag and swell.
4. Discuss the following characteristics of power quality events
 - a) Short duration variations
 - b) Long duration variations
 - c) Discuss in detail about transients
5. a) Explain the process of locating harmonic sources.
b) Explain the harmonic sources from commercial and industrial loads.
6. a) Discuss the harmonic effects on transformer and AC motors.
b) Explain power factor correction switching transients.
7. a) Define the term Harmonic distortions. Describe individual and total harmonic distortion.
b) What are the guidelines issued by IEEE for harmonic voltage and current limitations?
8. Write a short note on any two of the following
 - a) Electromagnetic compatibility
 - b) Variable tolerance band control
 - c) Data Loggers and Chart Recorders
 - d) Passive filters
